SYSTEM OVERVIEW

Jarvis Phase 2 — Cognitive Operating System Architecture

14 ADAPTERS · 8 SPECIALISTS · 3 MEMORY TIERS

INTERFACE LAYER

LLM ROUTER
gemma3:27b — Ollama

RICH CLI
main.py

FASTAPI
server.py — REST+WS

ANDROID
jarvis-android/

DAILY BRIEF
brief.py — TTS

CONSOLIDATION
(3 AM nightly)

MEMORY HIERARCHY

TIER 1
Working Memory
In-memory · 20 msgs
< 1ms · Session scope

TIER 2
Episodic Memory
SQLite · Full episodes
< 5ms · Weeks-months

TIER 3
Semantic Memory
ChromaDB + SQLite
< 50ms · Permanent

BONUS
Procedural Memory
Compiled patterns
Skip LLM for known Q

MEMORY BUS
Unified I/O — All tiers

KNOWLEDGE LAKE
facts.db + chromadb/ + kb_index

SPECIALIST LOOPS — BACKGROUND AI AGENTS

GROCERY specialist

FINANCE specialist

CALENDAR specialist

HOME specialist

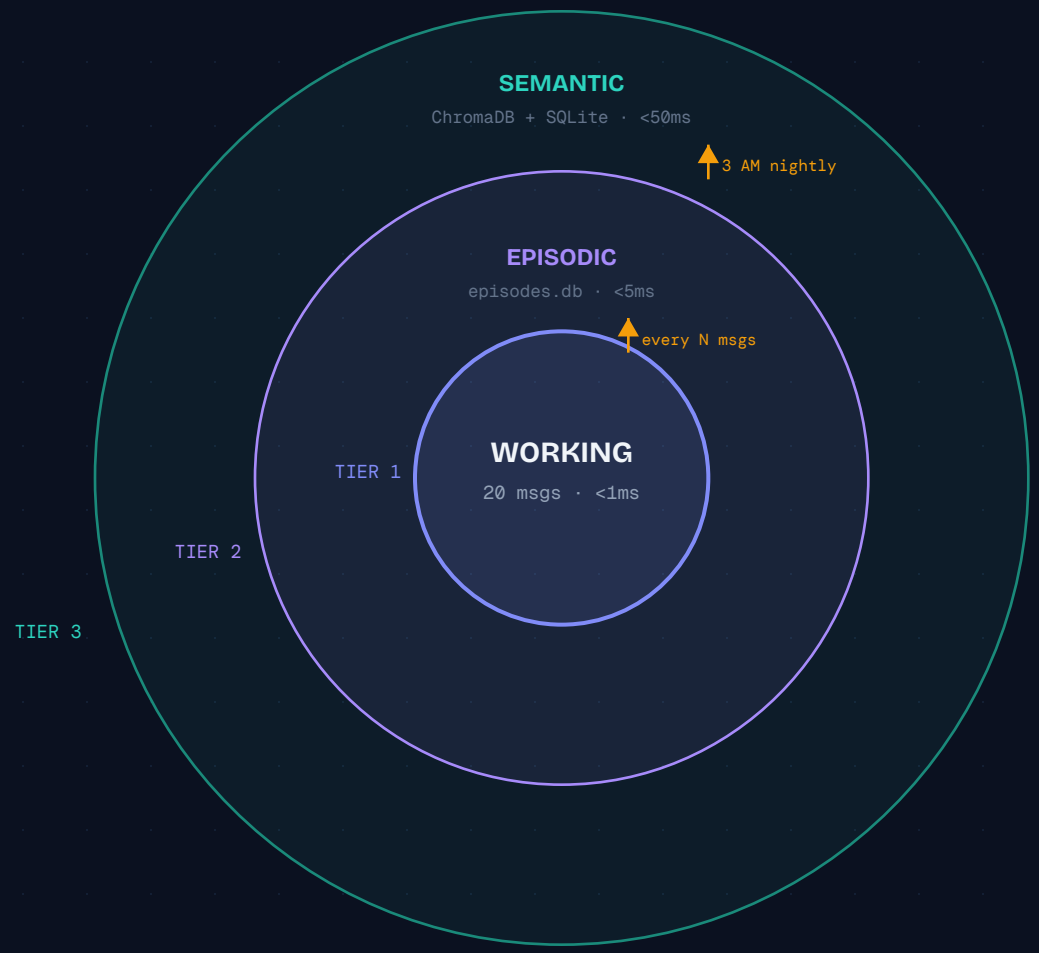
NEWS specialist

INVESTOR specialist

META specialist

MEMORY ARCHITECTURE

Brain-Inspired Three-Tier Hierarchy with Consolidation



CONSOLIDATION ENGINE

"Sleep Cycle" — Nightly at 3 AM

01 Replay unconsolidated episodes
02 Extract patterns via LLM
03 Merge into semantic store
04 Mark episodes consolidated
05 Prune low-value old episodes

ATTENTION GATE

Relevance-weighted prompt injection

0.35	Semantic similarity
0.20	Recency
0.15	Access frequency
0.15	Domain alignment
0.15	Surprise value

PROVENANCE CHAIN

Full audit trail for every fact

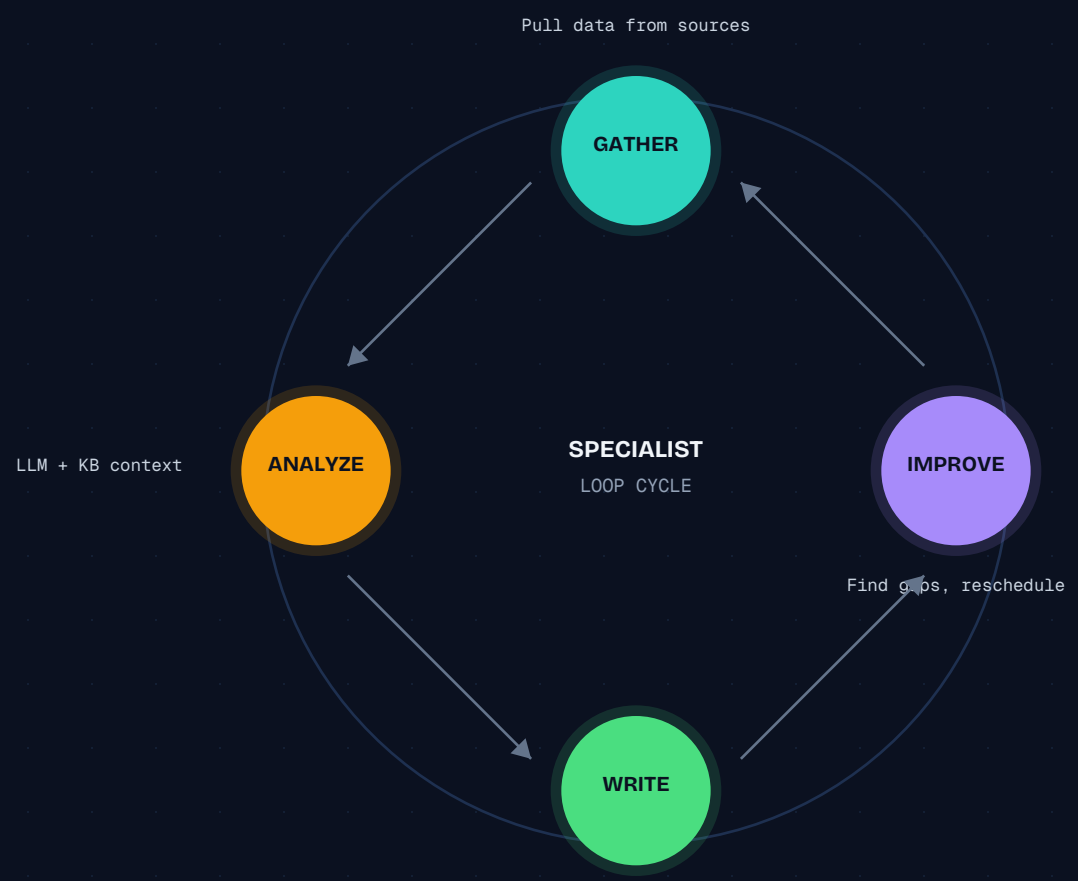
- source_type
- model_used
- prompt_hash
- confidence
- agent

ZETTELKASTEN LINKING (A-MEM, NeurIPS 2025)

Every knowledge note is atomic, richly tagged, and dynamically linked. New notes evolve existing ones.

SPECIALIST LOOP LIFECYCLE

Background AI Agents — Gather / Analyze / Write / Improve



SPECIALIST REGISTRY

- **GrocerySpec** — Every 4h
  Prices, deals, meals
- **FinanceSpec** — Daily
  Budget, spending, savings
- **CalendarSpec** — Every 2h
  Schedules, conflicts
- **HomeSpec** — Daily
  Maintenance, seasonal
- **NewsSpec** — Every 6h
  Headlines, market news
- **InvestorSpec** — Every 4h
  Portfolio, risk
- **HealthSpec** — Daily
  Appointments, wellness
- **ResearchSpec** — On-demand
  Deep dives, gap filling
- **MetaCognitive** — Daily
  Watches all specialists

CROSS-SPECIALIST COMMUNICATION

Specialists communicate through the Knowledge Lake, not directly

```
GrocerySpec writes: chicken on sale $1.99/lb at Aldi
FinanceSpec reads:  updates projected monthly spend
CalendarSpec writes: birthday party Saturday
```

HOUSEHOLD STATE

Composite state affects all specialists

- normal
- school_year
- summer
- holiday
- guests
- crunch
- budget_tight

LIBRARY OF ALEXANDRIA

Self-Building Local Knowledge System — Research Wings & Catalog

GROCERY WING

RESEARCH LIBRARIAN

- recipes
- pricing
- nutrition
- deals
- USDA data

context_engine.md
guidelines.md
resources/

FINANCE WING

RESEARCH LIBRARIAN

- budgets
- tax rules
- savings
- spending
- forecasts

context_engine.md
guidelines.md
resources/

HOME WING

RESEARCH LIBRARIAN

- repairs
- maintenance
- energy
- contractors
- seasonal

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resources/

INVESTING WING

RESEARCH LIBRARIAN

- market data
- strategies
- risk models
- papers
- filings

context_engine.md
guidelines.md
resources/

DEV/ENG WING

RESEARCH LIBRARIAN

- papers
- repos
- frameworks
- tools
- patterns

context_engine.md
guidelines.md
resources/

SHARED CATALOG

catalog.db — Cross-referenced · Searchable · Versioned · research_queue for cross-wing requests

LIBRARIAN LIFECYCLE

SURVEY → **EVALUATE** → **CATALOG** → **CURATE**

SURVEY	EVALUATE	CATALOG	CURATE
Cast wide net	Score relevance	Extract insights	Retire outdated
APIs, papers, repos	Actionability check	Store + cross-link	Identify gaps

HISTORICAL CONTEXT ENGINE

Longitudinal records that grow richer over time

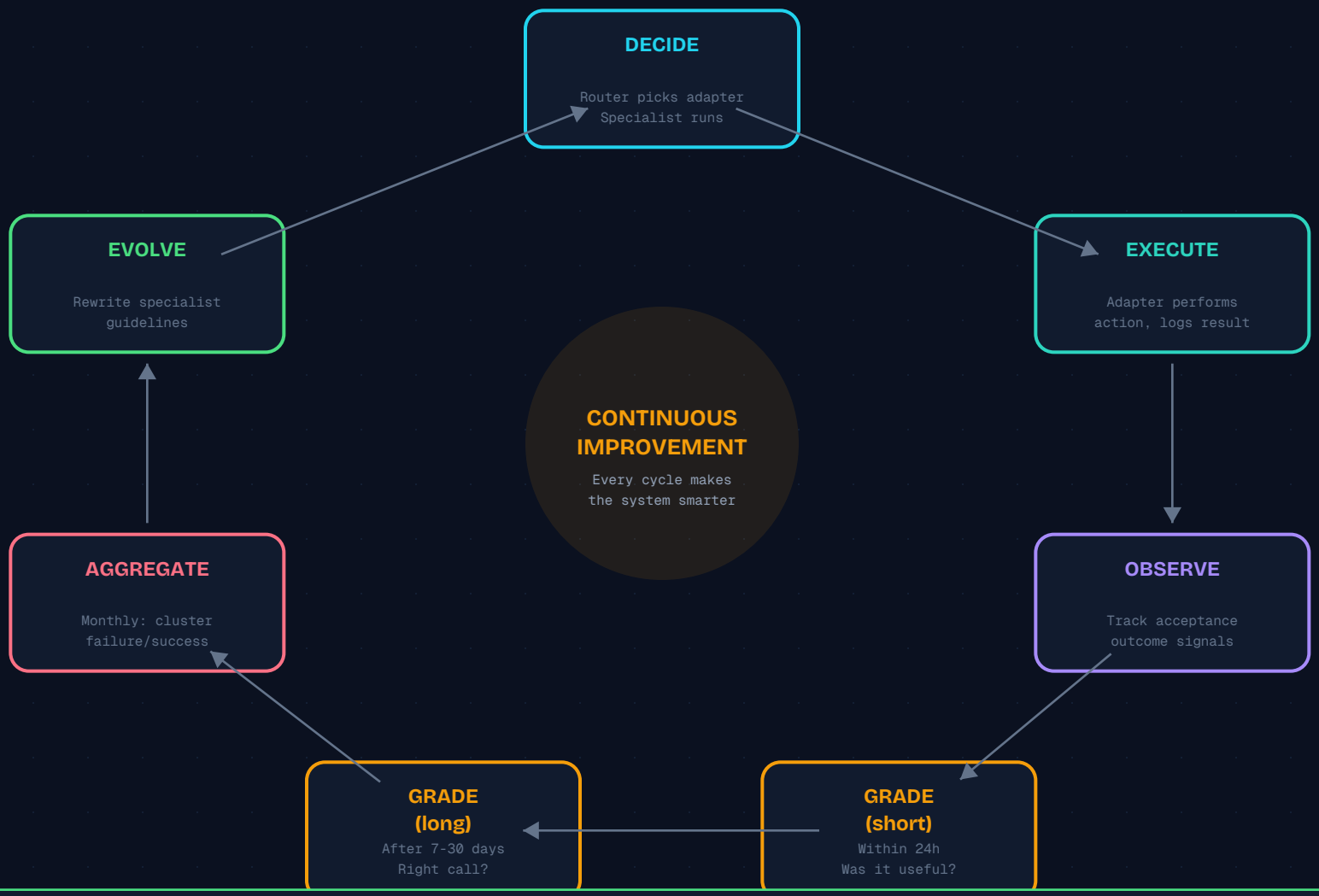
spending_history → price_history → budget_history

6 months: calibrated prediction models

12 months: seasonal pattern detection

SELF-IMPROVEMENT FEEDBACK LOOP

Decision Grading → Guideline Evolution → Context Engine Rebuild



GUIDELINE EVOLUTION EXAMPLE — GrocerySpec

v1 **3 rules**	v4 **7 rules**	v8 **10 rules**	v12 **12 rules**
Hand-written seed	+4 learned from 120 graded decisions	+3 seasonal patterns from summer data	+2 from school calendar integration

After 6 months: guidelines are battle-tested through 600+ graded decisions. The system wrote its own manual.